# Same Direction, Different Noise

A research report on rollout collection, gradient noise, and negative reinforcement in reinforcement learning with verifiable rewards

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Working report, October 18, 2026. Covers everything done so far; items marked **[running]** are still in progress.

## Summary

**The question.** When a language model is trained with reinforcement learning on math problems, it writes several answers per question, each answer is graded right or wrong, and the model is updated. Every design choice in this loop (how many answers to write, which answers to push up or down, and by how much) changes three things at once: which direction the update points, how noisy it is, and how much it costs. We set out to make these effects exact and measurable.

**What we found, in order.**

1. **An exact accounting.** For right-or-wrong rewards, the average update and the noise of *any* answer-collection rule depend on only four numbers per question. This holds even when the number of answers per question is random. The average part for fixed groups is known; the noise part with separate gradient sizes for right and wrong answers is the new piece.
2. **A new measurement.** In Qwen2.5-Math-1.5B, wrong answers have about 2.5 to 3 times larger squared gradients than right ones, while being only 9% longer. We found no earlier measurement of this, and two 2026 papers assume the opposite.
3. **A null result that taught us something.** Using that measurement to scale down wrong answers did not improve training (minus 2.2 and minus 2.7 points in two seeds). Smarter answer-collection rules gave small but consistent gains at equal cost: plus 1.1 and plus 3.3 points for our optimal rule on 3,382 held-out problems, plus 2.0 and plus 0.8 for probe-then-fill.
4. **The main finding.** Rewarding only right answers (PSR) and penalizing only wrong answers (NSR) have *exactly the same average update* (a direct consequence of the classical fact that a constant baseline does not change the expected policy gradient). Penalizing wrong answers equals rewarding right answers plus a term whose average is zero, pure noise. Because wrong answers carry larger gradients, NSR is about 6 times noisier (theory about 7, measured 6.4). In our experiments NSR learns more slowly, ends with lower pass@1, and reaches the *same* pass@64 as PSR and Dr. GRPO (83.0, 83.2, 83.2 in seed 1). At our scale this does not support the explanation in a NeurIPS 2025 paper that negative samples are special for preserving diversity; that paper used 4B to 8B models and $k$ up to 256, so our 1.5B result does not overturn it.
5. **Our noise formula holds in the real model.** The measured gradient noise matches the theorem within 30% in 12 of 16 runs, including all eight main runs.

**Where we are.** All eight pre-registered experiments are finished. For every way we tried of changing the update (penalizing wrong answers, five baselines in between, the NSR paper's

implementation, and even aiming the update at pass@4 instead of pass@1), we detected no shift of the accuracy-versus-diversity curve beyond about 2 points, also when reasoning is graded (CoT-Pass@k). These were point comparisons on 128 problems with two seeds, so they show no detected separation, not a statistically established equivalence; Pre-registrations 9 and 10 test equivalence with intervals. What differs is the speed along the curve, and the speed follows the noise our theorem predicts; whether noise *causes* it is tested by an intervention in Pre-registration 10. Open: generality beyond one 1.5B model and 150 steps (Pre-registration 9, running).

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# 1 Background: what we started from

**Reinforcement learning with verifiable rewards (RLVR).** For each math question, the model samples a group of $n$ answers. A program checks each final answer (reward 1 if right, 0 if wrong). The model then increases the probability of some answers and decreases others, weighted by an *advantage* $A_i$ per answer. The update is $g = \sum_i A_i \nabla \log \pi(y_i)$.

**GRPO and Dr. GRPO.** GRPO sets $A_i = (r_i - \bar{r})/\mathrm{std}(r)$ and averages each answer's loss over its tokens. Dr. GRPO ("GRPO done right") removes both divisions, which bias training, so $A_i = r_i - \bar{r}$. Example: if 2 of 8 answers are right, $\bar{r} = 0.25$, right answers get $+0.75$ and wrong answers get $-0.25$. Dr. GRPO is our main baseline.

**Adaptive answer collection.** Many recent methods decide adaptively how many answers to write per question: DAPO drops questions whose answers all agree and refills the batch; probe-then-fill methods (SPEED-RL, SARA, Pilot-Commit) write a few answers and continue only on questions with mixed results; others sample until a correct answer appears. These rules change the update's direction, noise, and cost at the same time, and they are usually compared by heuristics.

**The original idea (WORTH).** Make the value of each generated answer exact: derive the direction, noise, and cost of any collection rule, find the optimal rule, and test the predictions in real training.

# 2 Theory

## 2.1 The two-moment theorem

Let $p$ be a question's success probability, $\ell(y) = \nabla \log \pi(y)$ the score of answer $y$, $F_1$ and $F_0$ the mean squared score norms of right and wrong answers, and $\kappa(1) = 1/p$, $\kappa(0) = -1/(1-p)$.

**Theorem 1.** *For any rule that decides when to stop sampling and which advantages to assign from the rewards seen so far (including random group sizes),*

$$\mathbb{E}g = c(p)\,\nabla p, \qquad \mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p),$$

*where $c$, $D_1$, $D_0$, $X$ and the expected cost $\mathbb{E}N$ depend only on the rule and $p$.*

**Why it is true.** A right answer's average score is $\nabla p/p$ and a wrong answer's is $-\nabla p/(1-p)$. Given the whole sequence of rewards, the stopping time and advantages are fixed numbers and the answers are independent, so averaging over answers and then over rewards gives both formulas. No optional-stopping argument is needed. **What is new:** the average update for fixed groups was known (Davis and Recht, 2025; ARCUS, 2026), and invariance of the expected gradient to a constant baseline is classical (Williams, 1992; Greensmith, Bartlett and Baxter, 2004); the second moment with separate $F_0$ and $F_1$ and random group sizes is not in the literature we found. The derivation is short; the generality and the measured $F_0 \neq F_1$ are the contribution. All statements are verified by exact enumeration in small softmax models (`theory/exact.py`).

## 2.2 Efficiency and the optimal rule

Treating questions as having unrelated gradients and choosing the step size optimally, the progress in mean accuracy per generated answer is proportional to $\mathrm{Sig}^2/(\mathrm{Cost} \cdot \mathrm{M2})$. Dynamic programming

over the lattice of (wrong, right) counts finds the optimal rule. Main conclusions, exact within the model:

- DAPO's dynamic sampling has exactly the per-answer efficiency of the fixed group it filters, because dropped questions cost their answers and contribute nothing.
- Sample-until-correct rules are the least efficient.
- Probe-then-fill is the best rule in use; the optimal rule beats it by 1.3 to 1.7 times, and Dr. GRPO by 1.7 to 2.3 times, with the measured inputs.

| **Rule** | **Uniform** | **Mostly hard** | **U-shaped** | **Mostly easy** |
|---|---|---|---|---|
| Fixed group of 8 (GRPO, Dr. GRPO, DAPO) | 0.61 | 0.38 | 0.40 | 0.38 |
| Fixed group of 16 | 0.65 | 0.41 | 0.43 | 0.41 |
| Probe 4, abandon if all agree, fill to 16 | 0.77 | 0.60 | 0.67 | 0.60 |
| Sample until 1 correct, cap 16 | 0.26 | 0.09 | 0.12 | 0.30 |

Table 1: Efficiency relative to the optimal rule, by the difficulty mix of the questions.

### 2.3 When does a better rule matter? The critical budget

Without the assumption that questions have unrelated gradients, the exact one-step progress at $B$ answers per step is $\langle \nabla J, \mu \rangle^2 / (2L(s + nV/B))$, with $\mu$ the mean update per question, $s = \|\mu\|^2$, and $V$ its noise. The *critical budget* $B^\star = nV/s$ separates two regimes: below it, a rule with $k$ times the efficiency is worth $k$ times the answers (about $\sqrt{k}$ under Adam at a fixed learning rate); above it, no rule helps (Figure 1). We later found that this framing is already published for GRPO (Bansal et al., HiLD at ICML 2026), so it is now background, not a contribution.

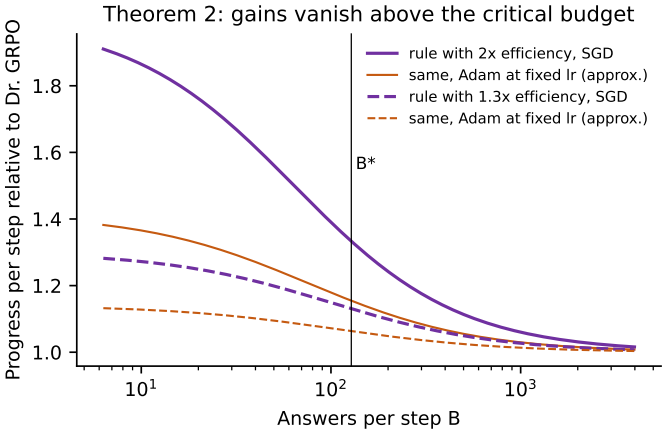



Figure 1: Predicted gain of a better rule against answers per step. Gains vanish above the critical budget, and Adam at a fixed learning rate turns a gain of $k$ into about $\sqrt{k}$.

### 2.4 Positive and negative reinforcement: same mean, different noise

Take advantages that depend only on an answer's own reward: $a$ for right, $b$ for wrong. The theorem gives

$$\mathbb{E}g = n(a-b)\,\nabla p, \qquad \text{noise} \approx n\big(a^2 pF_1 + b^2(1-p)F_0\big).$$

- **PSR** ($a=1$, $b=0$) and **NSR** ($a=0$, $b=-1$) both have mean $n\nabla p$: *exactly the same expected update*. This is baseline invariance, not a new result; what is new is the noise accounting below and its measurement.
- In fact NSR = PSR $-\sum_i \nabla \log \pi(y_i)$. The extra term has expected value exactly zero (shifting every advantage by a constant), so it is pure noise.
- PSR's noise is $n\,p\,F_1$ (it lives on easy questions); NSR's is $n(1-p)F_0$ (on hard questions). With $F_0/F_1 \approx 2.5$, NSR is noisier on every question solved less than 71% of the time.
- The whole family $A = r - b$ shares the mean, with noise $n[(1-b)^2 pF_1 + b^2(1-p)F_0]$, smallest near $b^\star = pF_1/(pF_1 + (1-p)F_0) \approx 0.15$, well below the group mean that Dr. GRPO and RLOO use.

**Consequence.** To first order in the step size, any difference between PSR and NSR (entropy, pass@k, stability) must come from the noise, from an optimizer that reacts to noise (Adam), or from implementation choices outside the theorem (per-token averaging, clipping, stale samples).

## 3 Measurements in a real model

### 3.1 Wrong answers carry larger gradients

We computed exact per-answer gradients in Qwen2.5-Math-1.5B on MATH training problems.

- Last-layer block, 40 questions by 16 answers: $F_1/F_0 = 0.36$ (90% intervals do not overlap), while wrong answers are only 9% longer (421 vs 386 tokens).
- All 1.5B parameters, calibration before each training run: $F_1/F_0$ between 0.32 and 0.42 across runs, so wrong answers carry 2.4 to 3.1 times larger squared gradients.
- Likely reason: wrong answers contain more low-probability tokens, whose scores are large.

This contradicts Assumption B.3 of VSR (Wang, Liu, Wang, October 2026: "the expected squared score-function norm does not depend on the realized reward") and an approximation in BICC (2026).

### 3.2 Gradient signal against difficulty

With 96 answers per question in the last-layer MLP, the unbiased estimate of $\|\nabla p\|^2$ is $0.73 \pm 0.60$ on very easy or very hard questions and $15.5 \pm 5.4$ on medium ones (Figure 2). The signal fades at both ends, best fit $p^2(1-p)^2$.

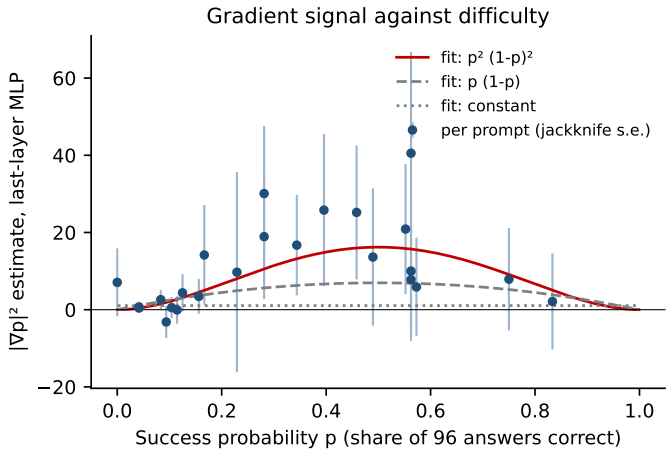



Figure 2: Per-question gradient signal against success probability, with fits of three shapes.

### 3.3 Measuring noise inside training

At every training step we split the questions into two random halves and record the inner products of the two halves' gradients and of each half with an independent unbiased reference for the population gradient. This gives, per step and without extra sampling, the mean update's size, its noise, and the noise scale. Two results:

- **The noise formula holds in the real model.** For every rule, the measured per-question second moment divided by the theorem's prediction is between 0.68 and 1.00 (Figure 3), inside the pre-registered 30% tolerance for all eight main runs.
- **Training becomes more noise-limited as it plateaus.** Dr. GRPO's noise scale grows from about 900 answers (steps 0 to 49) to over 10,000 (steps 50 to 149) in both seeds: the shared signal shrinks while the noise does not.

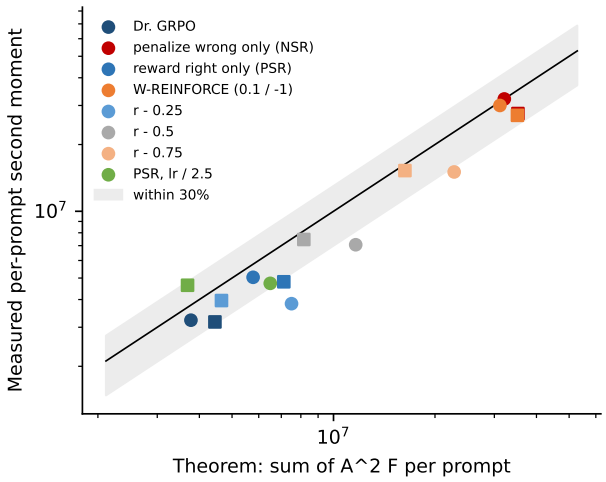



Figure 3: Measured per-question gradient second moment (steps 0 to 49) against the theorem's prediction from $F_0$, $F_1$ and the observed reward counts. Circles: odd seeds; squares: even seeds. Shaded: within 30%. The dose-response runs (B25, B50, B75, PSRlow) have few steps so far.

# 4 Experiments

All experiments use Qwen2.5-Math-1.5B base, 3,920 numeric MATH training problems, strictly on-policy updates with sequence-sum losses, AdamW at $10^{-6}$, 16 questions by 8 answers per step (128 answers), and 150 steps, unless stated. Every experiment was pre-registered (`experiments/training/PREREGISTRATION_*.md`) before its runs started. Evaluation: greedy accuracy on the 364 numeric MATH-500 problems, sampled accuracy and pass@k, and in later runs a 3,382-problem held-out MATH set.

## 4.1 Pilot: scale down wrong answers (Pre-registration 1)

**Why.** With $F_0 > F_1$, the theory says to weight a wrong answer by about $F_1/F_0$, a one-line change to GRPO predicted to raise efficiency by 1.1 to 1.6 times. **Result.** No gain: 45.3% vs 47.5% (seed 1) and 44.0% vs 46.7% (seed 2). Training reward was slightly higher early, then tied (Figure 4). **Verdict: no evidence. Lesson.** Under Adam at a fixed learning rate, the predicted 1.1 to 1.6 times shrinks to 1.06 to 1.25 times, too small to see in two seeds.

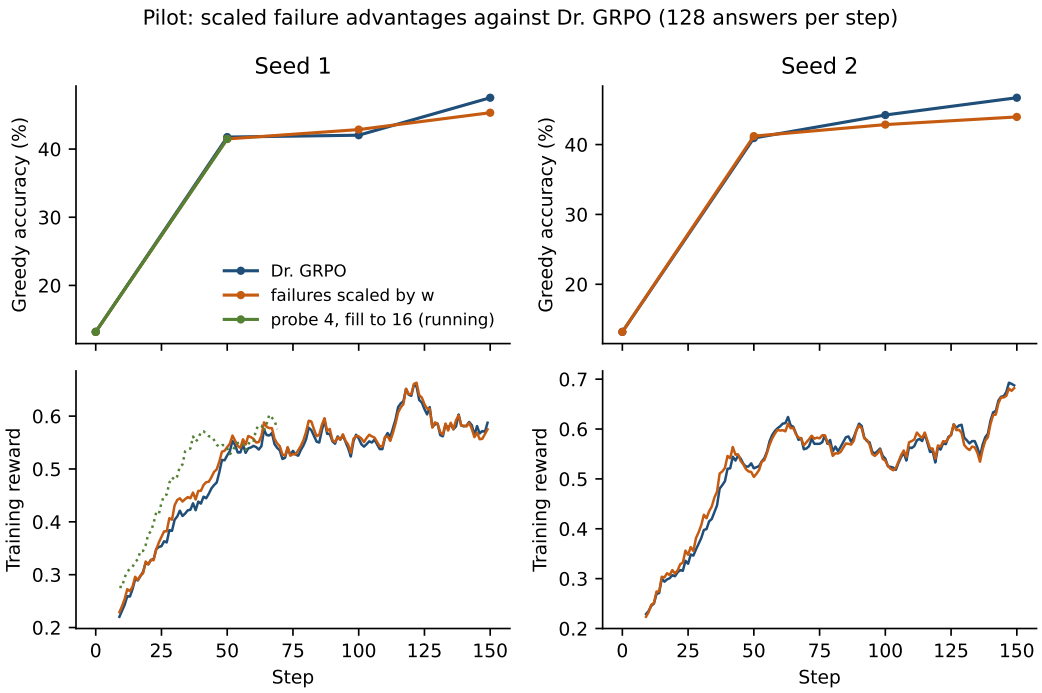



Figure 4: Pilot. Top: greedy accuracy. Bottom: training reward. Scaled failures (orange) track Dr. GRPO (blue). Probe-then-fill (green) is dotted because its training reward weights questions differently.

### 4.2 Answer-collection rules at equal cost (Pre-registrations 2 and 3)

**Why.** Test the optimal rule and probe-then-fill at exactly matched generated answers (19,200), with paired per-problem comparisons on the 3,382-problem set.

| **Run (versus Dr. GRPO at 128 answers per step)** | **Seed 1** | **Seed 2** |
|---|---|---|
| Optimal rule, 128 answers per step (paired, 3,382 problems) | $+1.1 \pm 0.4$ | $+3.3 \pm 0.5$ |
| Scaled failures (paired, 3,382 problems) | $-0.4 \pm 0.4$ | $+1.1 \pm 0.4$ |
| Dr. GRPO with 32 answers per step (paired, 3,382 problems) | $-4.4 \pm 0.5$ | not run |
| Probe-then-fill (MATH-500 greedy, against the pilot's Dr. GRPO) | $+2.0$ | $+0.8$ |

Table 2: Accuracy differences in points (standard errors for paired comparisons).

**Reading.** The optimal rule gains in both seeds, $+1.1 \pm 0.4$ and $+3.3 \pm 0.5$ (2.7 and 6.6 standard errors), with matched answers (19,172 and 19,200) and about 1% more tokens. The strict pre-registered rule (at least 1.5 points in both seeds) is not met because seed 1 falls short. Probe-then-fill is positive in both seeds. Gains are much smaller than the one-step theory predicts, consistent with Adam's square-root effect. Fewer answers per step clearly hurts (32 vs 128: minus 4.4 points).

### 4.3 Literature review and change of direction

Three literature reviews (verified by hand on the key papers) found:

- **The critical-budget framing is taken:** Bansal, Mohri, Abreu, Alvarez-Melis, Kakade, "Critical Batch Size for LLM Policy Optimization" (HiLD at ICML 2026) decomposes GRPO noise into between-question and within-question terms and finds the within-question term dominant, as our measurements do. Li et al. (arXiv 2608.29296) show square-root learning-rate scaling; IsoCompute (arXiv 2603.12151) studies answers per question.
- **Common flaws in adaptive-rollout papers:** gains plotted against training steps while spending 2 to 3 times more answers per step (for example, SARA's plus 6.8 points comes with 3.3 times the answers), learning rate not retuned, single seeds, 30-problem benchmarks, best checkpoints chosen on the test set, mechanisms ("lower variance") never measured.
- **We found no paper that measures gradient norms by correctness**, or that separates the mean and the noise of positive and negative reinforcement.

So we moved the main question to positive versus negative reinforcement, where our theorem and measurement are unique.

### 4.4 PSR, NSR, Dr. GRPO and W-REINFORCE (Pre-registration 4)

**Why.** Zhu et al. (NeurIPS 2025) report that training only on negative samples (NSR) preserves entropy and pass@k while positive-only training (PSR) collapses them, and recommend W-REINFORCE (weight 0.1 on right answers, $-1$ on wrong). Our theorem says PSR and NSR have identical average updates, so any difference must come from noise or implementation. **Design.** Four rules, two seeds, strictly on-policy, with in-training noise measurements, an on-policy entropy probe (the first four answers of every question, the same for all rules), and pass@k with 64 samples per problem at step 150.

| **Rule** | **Train acc., steps 25 to 49** seed 1 / seed 2 | **pass@1 at 150** seed 1 / seed 2 | **pass@64 at 150** seed 1 / seed 2 | **Entropy, 100 to 150** seed 1 / seed 2 |
|---|---|---|---|---|
| Dr. GRPO | 50.6 / 48.0 | 43.3 / 43.2 | 83.2 / 81.9 | 0.227 / 0.224 |
| PSR (reward right only) | 51.3 / 48.2 | 42.4 / 42.0 | 83.2 / 83.8 | 0.227 / 0.226 |
| NSR (penalize wrong only) | 33.6 / 36.4 | 37.7 / 39.1 | 83.0 / 82.1 | 0.251 / 0.251 |
| W-REINFORCE | 38.6 / 38.0 | 38.5 / 39.8 | 80.5 / 81.3 | 0.243 / 0.256 |

Table 3: Main comparison. The untrained model has pass@1 13.8 and pass@64 79.9.

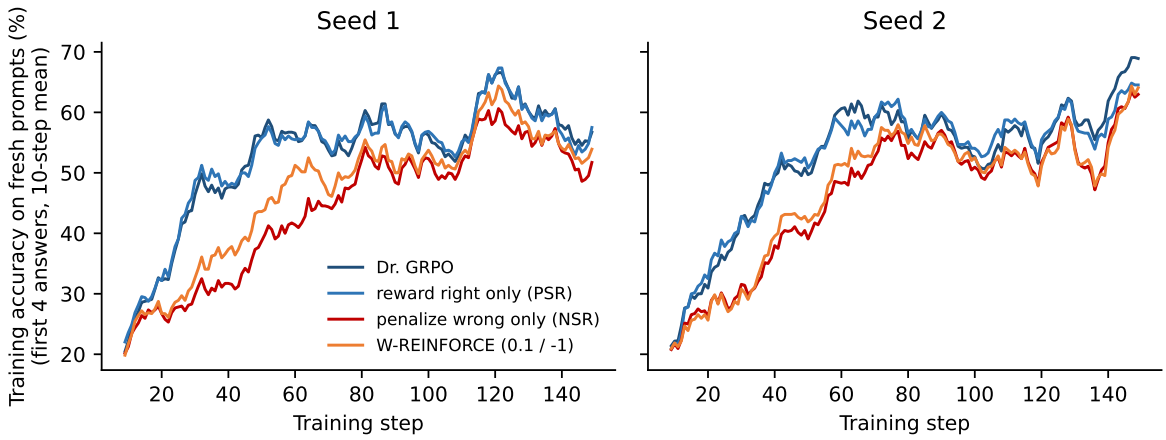



Figure 5: Training accuracy on fresh questions (unbiased for every rule). NSR and W-REINFORCE, whose updates are dominated by wrong answers, learn clearly more slowly.

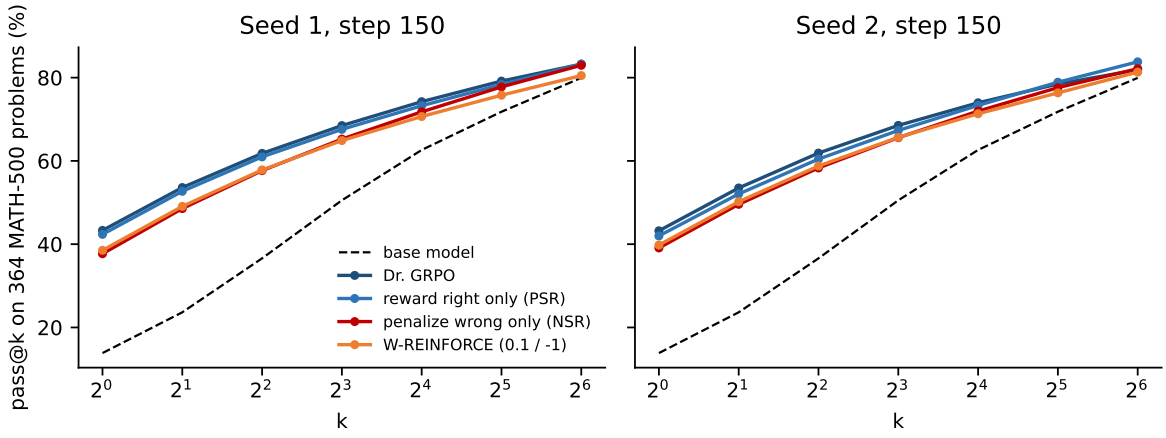



Figure 6: pass@k at step 150 against $k$. All rules converge at large $k$; NSR and W-REINFORCE are lower at small $k$.

**Results.**

- **Noise:** NSR's measured per-question noise is 6.4 times PSR's (theory about 7).
- **Speed:** NSR and W-REINFORCE learn more slowly (Figure 5) and end with 4 to 5 points lower pass@1.
- **Diversity:** pass@64 is the same for PSR, NSR and Dr. GRPO in seed 1 (83.2, 83.0, 83.2); W-REINFORCE is lower (80.5, 81.3). NSR's entropy is 11% higher than PSR's.
- **Equal-mean check:** the projection of each rule's mean update on the population gradient is too noisy to test (90% intervals include both 1 and 0).

**Verdict: the pre-registered separation (at least 3 points of pass@64 or 20% more entropy in favor of NSR) is not met in either seed: NSR minus PSR pass@64 is $-0.2$ and $-1.7$, and NSR's entropy is 1.11 times PSR's in both seeds.**

### 4.5 Equal accuracy, equal diversity? (Pre-registration 5)

**Why.** If NSR is just a slower PSR, its higher entropy reflects less progress. The decisive test compares diversity at the *same* pass@1: for each NSR checkpoint, PSR's pass@32 at the same pass@1 is interpolated along PSR's trajectory. **Result** (pass@32 on 128 problems, noise about ±2 to 3 points per point): NSR is on average 2.1 points *below* PSR's frontier in seed 3 and 0.2 points below in seed 4. **Verdict: no extra diversity at equal accuracy (Figure 7, left).**

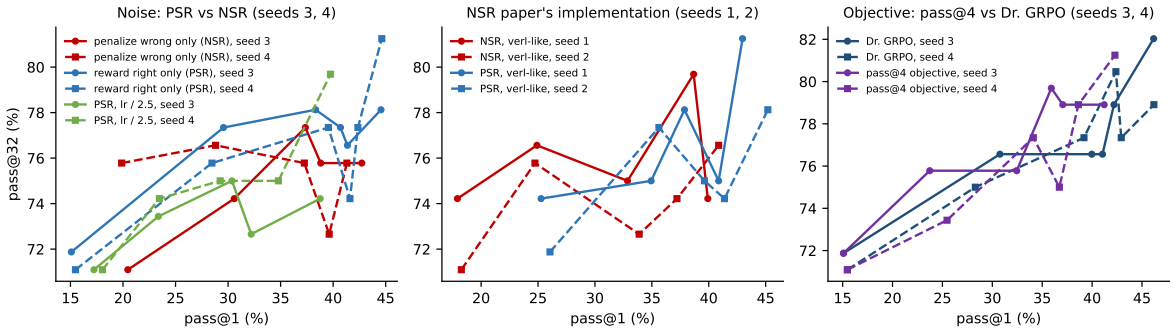



Figure 7: Accuracy against diversity during training (pass@1 against pass@32, 128 problems), at steps 0 to 150. Left: changing the noise (PSR, NSR, and PSR with a lower learning rate). Middle: the NSR paper's implementation choices. Right: changing the objective (pass@4 against Dr. GRPO). All runs trace the same curve.

### 4.6 The NSR paper's implementation choices (Pre-registration 6)

**Why.** Zhu et al. used verl, which differs from our setting in three ways the theorem does not cover: each rollout batch is reused for 4 updates with PPO clipping (stale samples), each answer's loss is averaged over its tokens, and PSR and NSR average over only the answers they train on. We reproduce all three at matched answers and updates; with fresh samples our clipped loss equals the plain loss exactly (difference 0.0). **Result.** NSR learns even more slowly in this setting (training accuracy 31.0 and 31.4 vs 43.0 and 45.4 over steps 25 to 49). Disclosure: per-token averaging makes gradients larger, so gradient clipping at norm 1 (as in verl's default) was active on 26 to 28% of NSR's steps and 7 to 8% of PSR's here; it was never active in any other on-policy run (largest norm 0.48). At step 150, NSR's pass@32 is 7.0 and 1.6 points *below* PSR's, and its entropy is 1.11 and 1.18 times PSR's. At matched pass@1 the differences are +0.3 and −1.3 points (Figure 7, right). **Verdict: not recreated: the NSR paper's implementation choices do not produce its diversity advantage at this scale.**

### 4.7 Dose-response across baselines (Pre-registration 7)

**Why.** The strongest test of "only noise differs": advantages $A = r - b$ for $b = 0, 0.25, 0.5, 0.75, 1$ share the mean exactly, while the theory predicts different noise. If outcomes follow the predicted noise, noise is the cause. A PSR run with the learning rate divided by 2.5 tests the "slower clock" reading. Verified: $b = 0$ reproduces PSR and $b = 1$ reproduces NSR exactly.

| | $b = 0.25$ | $b = 0$ (PSR) | $b = 0.5$ | $b = 0.75$ | $b = 1$ (NSR) | PSR, lr /2.5 |
|---|---|---|---|---|---|---|
| Predicted noise vs PSR | 0.81× | 1× | 1.25× | 2.5× | about 6× | 1× |
| Steps to 45% training accuracy, seed 3 | 37 | 44 | 44 | 53 | 56 | 84 |
| Steps to 45% training accuracy, seed 4 | 38 | 33 | 41 | 45 | 56 | 76 |
| pass@32 vs PSR at equal pass@1, seed 3 | +0.0 | 0 | +0.8 | −0.8 | −2.1 | |
| pass@32 vs PSR at equal pass@1, seed 4 | +0.2 | 0 | +2.1 | +0.6 | −0.2 | |

Table 4: Same average update, graded noise.

**Results.**

- **Noise predicts speed:** the learning-speed order follows the predicted noise in both seeds (rank correlation about 0.9 to 0.97; pre-registered threshold 0.8). **Verdict: passed.**
- **Noise formula:** measured within 30% of the theorem for all seed-4 runs (0.85 to 1.26), but only 0.51 to 0.73 for the seed-3 runs, so 12 of 16 runs pass overall. The likely cause is that gradient sizes are measured once before training and shrink as the model sharpens; the ordering is unaffected.
- **Mechanism:** every run stays within about ±2 points of PSR's curve at equal accuracy; "exploration" (noisy rules above the curve) is rejected. The strict "slower clock" rule asked for every difference below 2 points and two values are 2.07 and 2.14, so it is formally inconclusive. PSR with a 2.5 times lower learning rate learns *more* slowly than NSR, so NSR is equivalent to a smaller slowdown, and at equal accuracy it is indistinguishable from NSR (+0.6, −0.8).

### 4.8 Changing the objective: pass@4 (Pre-registration 8)

**Why.** If noise only changes speed, perhaps changing *what the update aims at* moves the curve. Unbiased pass@k objectives are known (PKPO, Walder and Karkhanis, 2025); here pass@4 serves as a control, not a contribution. We use an update aimed exactly at pass@4: per-answer weights from the unbiased pass@k estimator with a leave-one-out baseline, verified exactly over all 256 reward patterns to have average $2 \cdot 4(1-p)^3 \nabla p$, the pass@4 gradient. It pushes hard questions and ignores ones the model already solves. **Result.** At equal pass@1, pass@4 training is +1.6 (seed 3) and +0.2 (seed 4) points of pass@32 above Dr. GRPO, below the pre-registered +2. At step 150 it simply has lower pass@1 (41 to 42 vs 46). **Verdict: curve fixed.**

**Reasoning check (CoT-Pass@k).** Claude Haiku 5.5 graded every correct answer's reasoning three times (majority vote):

| Step 150, seed 3 / seed 4 | pass@1 | CoT-Pass@1 | pass@32 | CoT-Pass@32 |
|---|---|---|---|---|
| Dr. GRPO | 46.2 / 46.2 | 43.4 / 43.3 | 82.0 / 78.9 | 74.2 / 70.3 |
| pass@4 objective | 41.2 / 42.3 | 38.6 / 39.3 | 78.9 / 81.2 | 72.7 / 72.7 |
| Untrained model | 15.1 | 13.5 | 71.9 | 66.4 |

Reasoning-checked diversity is the same (CoT-Pass@32 72.7 vs 72.3 on average). About 93% of trained models' correct answers have valid reasoning, up from 90% before training, for both objectives. Judge cost: about $9.

# 5 Discussion

**What we have learned.**

1. In clean on-policy training, how you score right and wrong answers matters mainly through *noise*, and the noise is predictable from four numbers per question.
2. Wrong answers dominate the noise, because their gradients are 2.5 to 3 times larger.
3. Penalizing wrong answers alone does not preserve diversity in our setting; it learns more slowly and reaches the same pass@64.
4. **No rule we tried detectably moved the accuracy-versus-diversity curve:** noise, the NSR paper's implementation, and even a pass@4 objective changed mainly the speed along it (all differences within about $\pm 2$ points at matched pass@1, also when reasoning is graded). This is absence of detected separation, not yet a formal equivalence.
5. Smarter answer collection helps at equal cost: our optimal rule gains 1.1 and 3.3 points in two seeds, less than one-step theory predicts. Adam's response to noise is the likely reason the gain is smaller.

**How confident should we be?**

- **Strong:** the accounting (exact); the gradient asymmetry (all runs); NSR's slower learning; no extra diversity from NSR in any setting we ran, including the NSR paper's implementation.
- **Supported but correlational:** noise predicts learning speed (dose-response, two seeds). The baseline family changes the noise and the push on wrong answers together, so cause is not yet shown; Pre-registration 10 separates them.
- **Moderate:** the noise formula's absolute calibration (12 of 16 runs within 30%); the single curve, including the pass@4 objective (two seeds each, 128-problem point comparisons, no equivalence interval).
- **Open:** whether the single curve holds for larger models, longer training, and larger $k$.

**Threats to validity.**

- **One model family.** Qwen2.5-Math gains 21 points on MATH-500 even from random rewards (Spurious Rewards, 2025), so small effects on it are suspect; a second family is needed.
- **Short training.** 150 steps move pass@64 only from 79.9 to about 82 to 83, which leaves little room for differences.
- **Evaluation noise.** pass@32 on 128 problems carries about $\pm 2$ to 3 points; the final runs will use all 364 problems.
- **Judge reliability.** CoT-Pass@k depends on an LLM judge; it accepted correct and rejected flawed hand-written solutions, but was not validated against human grading at scale.
- **Theory scope.** One-step, SGD-based progress; the Adam statement is an approximation. Four numbers per question give the trace of the noise, not its covariance geometry or Adam's expected update, so equal means and calibrated noise do not by themselves imply equal trajectories.
- **Scale relative to the NSR paper.** Zhu et al. used 4B to 8B models and $k$ up to 256; a failure to reproduce at 1.5B does not overturn their result.

**Engineering issues fixed along the way.** GPU out-of-memory during evaluation (gradients kept alive; fixed by freeing them after each update, training unchanged), crashes on long held-out prompts (batch size now capped by prompt length; the final-check number and saved models were lost for Pre-registration 4), Hugging Face rate limits on shared AWS addresses (model now loaded from our S3 copy), and a grader crash on a sampled `\boxed{1/0}` (fixed and tested). Each fix was verified locally before relaunch.

## 6 External review and next steps

**Review (October 18).** We asked an independent model (Codex) to review the project as an ICML reviewer and area chair. Verdict: *reject as it stands (4 out of 10), with a credible path to a main-track diagnostic paper*; about 6 out of 10 if the second model family establishes one curve with intervals. Main points, all accepted:

- **Novelty is narrower than first framed.** Equal means for PSR and NSR follow from baseline invariance; unbiased pass@k objectives exist (PKPO); Dr. GRPO, the accuracy-versus-coverage framing (Yue et al.) and CoT-Pass@k (Wen et al.) are prior work. What is ours: the second-moment accounting for adaptive rules with correctness-dependent gradient sizes, the measured asymmetry, and matched-accuracy tests of the NSR explanation.
- **Absence of separation is not equivalence.** The Qwen comparisons were point estimates on 128 problems with two seeds; the slower-clock rule was formally inconclusive and the optimal rule missed its threshold.
- **Noise predicts speed; that it explains NSR is not shown.** An intervention is needed.

**What we are doing about it.**

1. **Wording** (done in this report and the brief): baseline invariance credited, no "nobody measured" claims, absence of separation stated as such, scale relative to the NSR paper stated.
2. **Second model family with intervals** (Pre-registration 9, running): OLMo-2-1B-SFT, 300 steps, 1,024-token answers, Dr. GRPO, PSR, NSR, pass@4 and the optimal rule, two seeds, pass@16 at matched pass@1 on 256 stratified held-out problems with a bootstrap equivalence test (90% interval inside $\pm 2$ points).
3. **Noise-only intervention** (Pre-registration 10, under review): PSR plus random-sign noise sized each step to NSR's predicted second moment. The expected update is unchanged and the noise matches NSR's, but nothing pushes wrong answers down systematically. If it learns as slowly as NSR, the size of the noise explains NSR; if it learns like PSR, NSR's specific push matters. Plus NSR at twice the learning rate, two more PSR/NSR seeds (a four-seed equivalence test on Qwen), $F_0/F_1$ re-measured during training, and noise after Adam's preconditioning.
4. **Later, if the above holds:** a larger evaluation at a few overlapping accuracy levels from saved checkpoints, longer PSR/NSR runs, and one larger backbone if it fits on one L40S.

**The claim we would submit, if the tests pass:** *in on-policy RLVR, positive and negative reinforcement share their expected update; NSR's extra term is zero-mean noise concentrated on wrong answers, which carry larger gradients; the size of that noise, predicted from four numbers per question, sets how fast a run moves along an accuracy-versus-diversity curve that reward shaping does not move.* The optimal answer-collection rule is the practical consequence. WORTH-k (a noise-minimized pass@k objective) is dropped.

# A Pre-registrations and verdicts

| # | Question | Verdict |
|---|---|---|
| 1 | Scaled failure advantages beat Dr. GRPO? | No evidence ($-2.2$, $-2.7$) |
| 2 | Optimal rule and probe-then-fill at matched answers? | Probe $+2.0$, $+0.8$; optimal replaced by 3 |
| 3 | Budget ladder, in-training noise, optimal rule at 128 | Optimal $+1.1 \pm 0.4$, $+3.3 \pm 0.5$; rule not met |
| 4 | PSR vs NSR vs Dr. GRPO vs W-REINFORCE | Separation not reproduced (both seeds) |
| 5 | Equal diversity at equal pass@1 (seeds 3, 4) | No extra diversity ($-2.1$, $-0.2$) |
| 6 | NSR paper's implementation choices | Not recreated ($-7.0$, $-1.6$ pass@32) |
| 7 | Baseline dose-response and slower-clock control | Speed follows noise (passed); mechanism formally inconclusive |
| 8 | pass@4 objective vs Dr. GRPO | Curve fixed ($+1.6$, $+0.2$); CoT-Pass@32 equal |
| 9 | Second model family (OLMo-2-1B), intervals | Running |
| 10 | Noise-only intervention, learning-rate control | Under review |

# B Compute

About 118 GPU-instance hours on SageMaker jobs finished before October 17 afternoon, roughly $480 by our price estimates, plus about 30 runs since (a few hundred dollars more), and about $9 of Claude Haiku 5.5 judging. All code, pre-registrations, results, and figures are in the repository `github.com/mayman150/rlvr-adaptive`.